

Biology Practice Test 06 (FINAL GENETICS MARATHON)

Format: Official ISC Board Exam Simulator

Portion: Set 1 Midterm (GENETICS HEAVY)

Time Allowed: 3 Hours

Maximum Marks: 70 (Note: The official ISC Biology Theory paper is 70 marks. We are strictly matching the board format for accurate time management).

General Instructions:

- **CRITICAL RULE:** Answer ALL parts of a question. Do not ignore the second sentence of a prompt.
 - **NO BLANKS ALLOWED:** A -1 mock penalty applies for leaving any question blank.
 - **Formatting Rule:** You MUST underline genus and species separately when writing scientific names.
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SECTION A (20 Marks)

Answer all questions. Each question carries 1 mark.

Question 1: Which two scientists independently proposed the Chromosomal Theory of Inheritance? **Question 2:** Name the specific type of sex determination (karyotype) seen in grasshoppers. **Question 3:** Identify the inborn error of metabolism (genetic disorder) caused by the absence of the enzyme phenylalanine hydroxylase. **Question 4:** State the exact phenotypic ratio of the F_2 generation in a classic Incomplete Dominance cross (e.g., *Mirabilis jalapa*). **Question 5:** Name the term used to describe the genetic phenomenon where a single gene exhibits multiple phenotypic expressions. **Question 6:** Which scientist extensively used the fruit fly (*Drosophila melanogaster*) to establish the chromosomal basis of genetics? **Question 7:** Name the type of endosperm development where cell wall formation occurs immediately after every nuclear division. **Question 8:** Name the specific type of pollination where pollen is transferred to the stigma of another flower on the *same* plant. **Question 9:** What is the specific final step in the HIV replication cycle where newly formed virions exit the host macrophage? **Question 10:** What term is used to describe the physical distance between two genes on a chromosome, which determines their frequency of crossing over? **Question 11:** Name the specific cells in the human testis that synthesize and secrete androgens (testosterone). **Question 12:** What is the phenomenon of producing seeds without fertilization in plants called? **Question 13:** Name one specific copper-releasing Intra Uterine Device (IUD). **Question 14:** Which evolutionary scientist proposed the biogenetic law stating that 'Ontogeny recapitulates Phylogeny'? **Question 15:** Name one fungal genus responsible for causing Ringworm in humans. **Question 16:** Name the triploid structure formed by the fusion of a male gamete with the secondary nucleus in angiosperms. **Question 17:** What is the medical term for the surgical sterilization method involving the removal of a small part of the vas deferens? **Question 18:** Name the shape of the population growth curve obtained when environmental resources are unlimited. **Question 19:** Name one specific example of a Primary Lymphoid Organ in humans. **Question 20:** Which specific property of normal cells prevents them from growing uncontrollably upon touching other cells?

SECTION B (14 Marks)

Answer all questions. Each question carries 2 marks.

Question 21: Differentiate between Homozygous and Heterozygous individuals with respect to alleles.

Question 22: What is a Pedigree Analysis? State its major significance in human genetics. **Question 23:** Briefly describe the chromosomal mechanism of sex determination in humans. **Question 24:** Differentiate between Autosomal Dominant and Autosomal Recessive traits. Give one example of an autosomal recessive Mendelian disorder. **Question 25:** Differentiate between Artificial Insemination (AI) and Intra Uterine Insemination (IUI). Under what specific medical conditions are these techniques recommended? (*Do not skip the second part!*)

Question 26: State the five major evolutionary factors that are known to affect Hardy-Weinberg equilibrium in a population. **Question 27:** Explain the biological mechanism behind the use of *Bacillus thuringiensis* (Bt). Specifically detail how the toxin kills the insect.

SECTION C (21 Marks)

Answer all questions. Each question carries 3 marks.

Question 28: Explain Mendel's Law of Independent Assortment with the help of a suitable dihybrid cross (using seed shape and color in peas). **Question 29:** Describe the Chromosomal Theory of Inheritance. State three distinct parallels between the behavior of chromosomes and the behavior of Mendelian factors (genes) during meiosis. **Question 30:** Explain T.H. Morgan's concept of Linkage and Recombination. Why did the F_2 generation in his *Drosophila* dihybrid cross deviate significantly from the expected 9:3:3:1 ratio?

Question 31: What are Intra Uterine Devices (IUDs)? Classify them into three distinct categories and provide one example for each category. **Question 32:** Differentiate between Spermatogenesis and Oogenesis (provide three distinct points of difference). **Question 33:** Describe the embryological support for evolution proposed by Ernst Haeckel. Why was this proposal ultimately rejected by Karl Ernst von Baer? **Question 34:** Discuss the pathogenesis, two major symptoms, and the causative organism of Pneumonia in humans.

SECTION D (15 Marks)

Answer any ONE question from each of the three pairs below. Each question carries 5 marks.

Question 35: (Option A) A woman heterozygous for blood group A marries a man who is heterozygous for blood group B. (i) Using a Punnett square, work out the possible blood groups of their offspring and state their probabilities. (3 Marks) (ii) Explain the genetic principles of Co-dominance and Multiple Alleles demonstrated by the human ABO blood grouping system. (2 Marks) **OR (Option B)** What is Thalassemia? Classify it into Alpha and Beta types based on the affected globin chains. How does the genetic mechanism of Thalassemia fundamentally differ from that of Sickle-cell anemia? (5 Marks)

Question 36: (Option A) Trace the complete life cycle of the Human Immunodeficiency Virus (HIV) inside a human macrophage. Detail the entry, the specific viral enzymes involved, and the final exit mechanism of the virus. (5 Marks) **OR (Option B)** Draw a well-labeled diagram of a typical anatropous ovule of an angiosperm. Briefly describe the specific functions of the integuments, the nucellus, and the synergids. (5 Marks)

Question 37: (Option A) Define a Test Cross and explain its genetic significance. If a tall pea plant with purple flowers (heterozygous for both traits, $TtPp$) is test crossed with a homozygous recessive plant ($ttpp$), what will be the phenotypic ratio of the offspring? Show the cross using a Punnett square. (5 Marks) **OR (Option B)** Explain the various components of a typical Biogas plant with the help of a labeled diagram. Detail the role of methanogens in the anaerobic production of biogas. (5 Marks)